Year 8 Chemistry Homework

Name:	Due Date

Match the scientific vocabulary below to the definitions in the table.
Try to make sure you can spell and remember what each word means.

Atomic Number **Chemical Formula** **Chemical Symbol** **Compound**

Element **Molecule** **Relative Formula Mass**

Scientific vocabulary	Definition
	Substances that all other materials are made up of, and which contain only one type of atom. They cannot be broken down into other substances.
	Pure substances made up of atoms of two or more elements, strongly joined together.
	A group of two or more (up to thousands) atoms strongly joined together. Most non-metal elements exist either as small or giant versions of this.
	A one- or two-letter code for an element that is used by scientists in all countries.
	A formula that shows the elements present in a compound and their relative proportions.
	The number of protons (which equals the number of electrons) in an atom. It is sometimes called the proton number.
	The sum of the relative atomic masses of the atoms in the numbers shown in the formula.

Q1. The chemical formulae for four acids are shown in the table below.

sulphuric acid	hydrochloric acid	nitric acid	ethanoic acid
H_2SO_4	HCl	HNO_3	CH_3COOH

(i) Give the **name** of the element that is present in all four acids.

.....

(ii) Give the **names** of the two **other** elements present in sulphuric acid.

1. .....

2. .....

(iii) How many atoms are there in the formula HNO_3 (nitric acid)?

.....

Q2. Gemstones called rubies are made from an aluminium compound with the formula Al_2O_3.

The chemical symbol for aluminium is Al.

(i) Give the name of the element that is combined with aluminium in this compound.

.....

(ii) Suggest the name of the compound with the formula Al_2O_3.

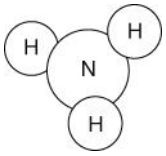
.....
.

(iii) How many atoms are there in the formula Al_2O_3?

.....

Q3. By counting up the atoms, write the chemical formula beside each of the molecules.

1



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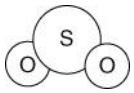
ammonia

2



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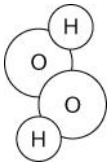
sulfur dioxide

3



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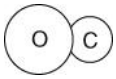
hydrogen peroxide

4



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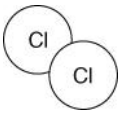
carbon monoxide

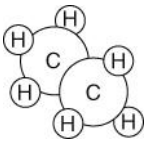
5



_____

chlorine

6



_____

ethane

Q4. Draw a diagram of a methane molecule below, formula CH_4.

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Match the scientific vocabulary below to the definitions in the table.
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Atom **Electron** **Isotope** **Mass Number**

Neutron **Nucleus** **Proton** **Shell**

Scientific vocabulary	Definition
	The smallest part of an element that can exist.
	A tiny positive particle found inside the nucleus of an atom.
	A dense particle found in the nucleus of an atom. It is electrically neutral, carrying no charge.
	A tiny particle with a negative charge. Electrons orbit the nucleus of atoms or ions in shells.
	The very small and dense central part of an atom that contains protons and neutrons.
	An area in an atom, around its nucleus, where electrons are found.
	Atoms that have the same number of protons but different number of neutrons, i.e., they have the same atomic number but different mass numbers.
	The number of protons plus neutrons in the nucleus of an atom.

Q1 What is the approximate radius of an atom? Tick (✓) **one** box.

0.1 m ☐ 0.1 mm ☐ 0.1 nm ☐

Q2 What is the charge of an electron? Tick (✓) **one** box.

–1 ☐

0 ☐

+1 ☐

Q3. **Figure 1** represents an atom of sulfur.

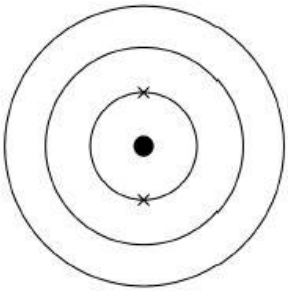
Figure 1

$^{32}_{16}S$

(a) Complete the table below.

Particle	**Number of particles in a sulfur atom**
Electron	16
Neutron	
Proton	

(b) Complete the electron shell diagram of the sulfur atom.



Q4.

(a) Which sub-atomic particles are present in the nucleus of an atom?

_____ and _____

(b) There are two isotopes of the element chlorine:

$^{35}_{17}Cl$ $^{37}_{17}Cl$

Describe, in terms of **sub-atomic particles**, **one** similarity and **one** difference between atoms of the two isotopes of chlorine.

Similarity

_____

Difference

_____

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Match the scientific vocabulary below to the definitions in the table.
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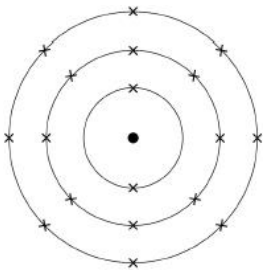
Chemical Properties **Group** **Noble Gases** **Period**

Periodic Table **Physical Properties** **Trend** **Unreactive**

Scientific vocabulary	Definition
	A table which shows all the elements arranged in columns and rows. Elements with similar properties are grouped together.
	A column of the Periodic Table. The elements in a group have similar properties.
	A row of the Periodic Table. There are trends in the properties of the elements across a period.
	Features of the way a substance reacts with other substances.
	Features of a substance that can be observed without changing the substance itself e.g. appearance
	A pattern in properties, such as an increase or decrease.
	The name for elements in the group on the right of the Periodic Table. Noble gases include helium, neon, argon, and krypton. Also known as the Group 0 elements.
	Elements that take part in few chemical reactions are unreactive.

Q1 Argon is very unreactive. The diagram represents the electronic structure of an argon atom.

(a) How does the electronic structure show that argon is unreactive?



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(b) What is the name of the group that contains argon? Tick (✓) **one** box

Alkali metals ☐

Halogens ☐

Noble gases ☐